2.2.7 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of generating equivalent algebraic expressions to find quotient of two monomials using the properties (laws) of exponents.

0 1 2 3 4 5 6 7 8 9 10

2. Complete the statement.

I chose this emoji because ______________________________

____

____

____

Unit 2, Lesson 3: Applying Exponent Laws to Generate Equivalent Algebraic Expressions

Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.

Learning Target

☐ I can generate equivalent algebraic expressions using properties (laws) of exponents.

2.3.1 Warm-Up: Dividing Monomials

1. Using the digits 1 to 10, at most one time each, fill in the boxes to make the statement true.

$$\frac{\square \cdot x^{\square} \cdot y^{\square}}{\square \cdot y^{\square}} = \frac{\square \cdot x^{\square} \cdot y^{\square}}{\square \cdot x^{\square}}$$

2.3.2 Guided Practice: Using Different Methods to Generate Equivalent Expressions

1. Consider the expression $\frac{(4x)(7x^3)}{2x^2}$.

   a. Use expanded form to generate an equivalent expression for $\frac{(4x)(7x^3)}{2x^2}$, where $x \neq 0$.

$$\frac{(4x)(7x^3)}{2x^2} = \frac{(4 \cdot \quad)(7 \cdot \quad \cdot \quad \cdot \quad)}{(2)(x \cdot \quad)} =$$

   b. For the expression shown, indicate the laws of exponents used to generate the equivalent expressions.

$\frac{(4x)(7x^3)}{2x^2} = \frac{28x^4}{2x^2} = 14x^2$	☐ Negative Exponent ☐ Product of Powers ☐ Quotient of Powers ☐ Power of a Product ☐ Zero Exponent ☐ Power of a Power ☐ Power of a Quotient

2. Three students used different methods to generate an equivalent expression for the expression $\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2$, where $a \neq 0$.

a. Finish the work for each student.

Sebastian	Ana
$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$ $\frac{25a^6 \cdot 9a^8}{36a^4} =$	$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$ $\left(\frac{15a^7}{6a^2}\right)^2 =$

Libby
$\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)^2 =$ $\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right)\left(\frac{5a^3 \cdot 3a^4}{6a^2}\right) =$ $\left(\frac{5 \cdot 5 \cdot 3 \cdot 3 \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a}{6 \cdot 6 \cdot a \cdot a \cdot a \cdot a}\right) =$

b. Discuss with your partner the differences in the methods used by each student. Then, determine if the different methods each produced the same equivalent expression.

2.3.3 Try It: Generating Equivalent Algebraic Expressions

1. For each of the following, find an equivalent expression.

$\dfrac{(-10h)^2}{5h^6}$ where $h \neq 0$	$\left(\dfrac{-2c \cdot -3c^4}{8c^2 \cdot -6c}\right)^2$ where $c \neq 0$	$\left(\dfrac{12ry^2 \cdot 3r^4y^5}{6r^7y^{10}}\right)$ where $r \neq 0$ and $y \neq 0$

2. Compare your answers with your partner and resolve any differences, if necessary.

3. Compare and discuss the methods you used to create each equivalent algebraic expression.

2.3.4 Bring It Together: Generating Multiple Equivalent Algebraic Expressions

1. Discuss with your partner how to show $3^{(2x+4)}$ is equivalent to $81(9)^x$. Summarize your discussion.

2. Prove $3^{(2x+4)} = 81(9)^x$ by filling in the value for each blank.

$$3^{(2x+4)} = 3^{(2x)} \cdot 3^{\boxed{}}$$

$$3^{(2x)} \cdot 3^{\boxed{}} = 3^{(2x)} \cdot 81$$

$$81 \cdot 3^{(2x)} = (3^2)^{\boxed{}} \cdot 81$$

$$(3^2)^{\boxed{}} \cdot 81 = (9)^{\boxed{}} \cdot 81$$

$$3^{(2x+4)} = 81(9)^x$$

3. Prove $z^{(5y-6)} = \frac{(z^5)^y}{z^6}$ by using the laws of exponents. Show your work.

4. Use the laws of exponents to write two equivalent expressions for $1.2^{(4+2x)}$. Show your work.

2.3.5 Your Turn!

1. Write an equivalent expression for $3^{9r^2} \cdot 3^{9r}$ using the laws of exponents.

2. Prove $2.1^{(3x+1)} = 2.1(9.261)^x$ by using the laws of exponents. Show your work.

3. Use the law of exponents to write two equivalent expressions for $6.2^{(2-3x)}$. Show your work.

2.3.6 Wrap-Up: Error Analysis

1. Errors were made in finding expressions equivalent for $4^{(3s-1)}$. Identify the errors, then write two correct equivalent expressions.

$$4^{(3s-1)} = \frac{4^{(3s)}}{4^1} = \frac{(4^3)^s}{4^1} = \frac{64 \cdot 4^s}{4} = 4 \cdot 4^s$$

Unit 2, Lesson 4: Introducing Polynomials

Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Targets

- ☐ I can write polynomials in standard form.
- ☐ I can multiply a monomial to a polynomial.